

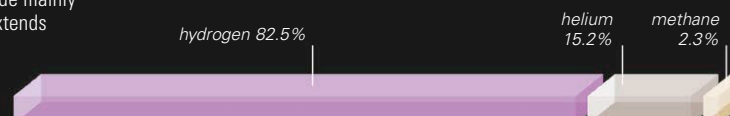
ATMOSPHERE AND WEATHER

Uranus's blue color is a result of the absorption of the incoming sunlight's red wavelengths by methane-ice clouds within the planet's cold atmosphere. The cloudtop temperature of -364°F (-220°C) appears to be fairly uniform across the planet. The action of ultraviolet sunlight on the methane produces haze particles, and these hide the lower atmosphere, making Uranus appear calm.

The planet is, however, actively changing. The Voyager 2 data revealed the movement of ammonia and water clouds around Uranus, carried by wind and the planet's rotation. It also revealed that Uranus radiates about the same amount of energy as it receives from the Sun and has no significant internal heat to drive a complex weather system. More recently, observations made using ground-based telescopes have also made it possible for astronomers to track changes in Uranus's atmosphere.

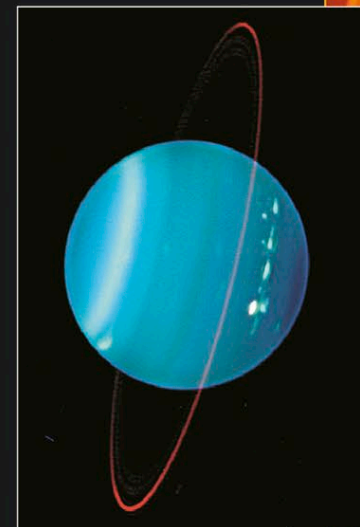
COMPOSITION OF ATMOSPHERE

The atmosphere is made mainly of hydrogen, which extends beyond the visible cloudtops and forms a corona around Uranus.



CLOUDS

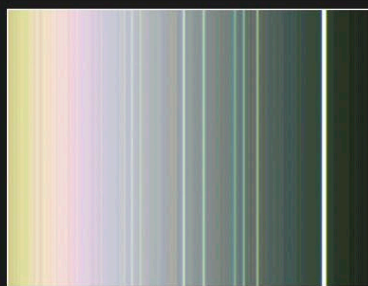
This Keck II telescope infrared image has been processed to show vertical structure. The highest clouds appear white; mid-level ones, bright blue; and the lowest clouds, darker blue. As a by-product, the rings are colored red.



RINGS AND MOONS

Uranus has 11 rings that together extend out from 7,700 to 15,900 miles (12,400–25,600 km) from the planet. The rings are so widely separated and so narrow that the system has more gap than ring. All but the inner and outer rings are between 0.6 and 8 miles (1 and 13 km) wide, and all are less than 9 miles (15 km) high. They are made of charcoal-dark pieces of carbon-rich material measuring from a few inches to possibly a few yards across, plus dust particles. The first five rings were discovered in 1977 (see panel, right). The rings do not lie quite in the equatorial plane, nor are they circular or of uniform width. This is probably due to the gravitational influence of small, nearby moons. One of these, Cordelia, lies within the ring system. Uranus has 27 moons. The five major moons were discovered using Earth-based telescopes. Smaller ones have been found since

the mid-1980s, through analysis of Voyager 2 data or by using today's improved observing techniques. More discoveries are expected.



FALSE-COLOR VIEW OF THE RINGS

Nine of Uranus's rings are visible in this Voyager 2 image. The faint pastel lines are due to image enhancement. The brightest, colorless ring (far right) is the outermost ring, epsilon. To its left are five rings in shades of blue-green, then three in off-white.

EXPLORING SPACE

RINGS DISCOVERED

In March 1977, astronomers onboard the Kuiper Airborne Observatory, an adapted high-flying aircraft, were preparing to observe a rare occultation of a star by Uranus in order to measure the planet's diameter. Before the star was covered by the planet's disk, it blinked on and off five times. A second set of blinks was recorded after the star appeared from behind the planet. Rings around Uranus had blocked out the star's light.

KUIPER AIRBORNE OBSERVATORY

Astronomers and technicians operate an infrared telescope, which looks out to space through an open door in the side of the aircraft.



URANUS'S MOONS

